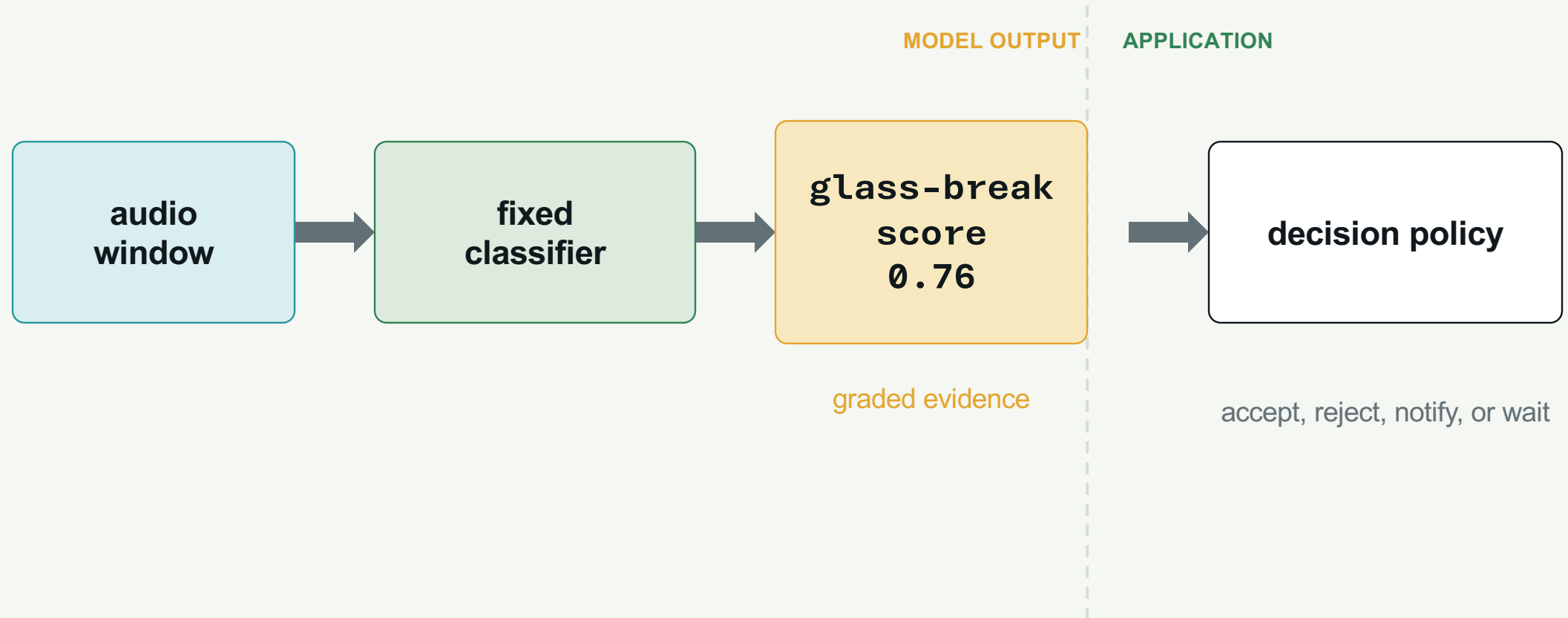


# Classification Thresholds and Operating Points

Turning model scores into decisions

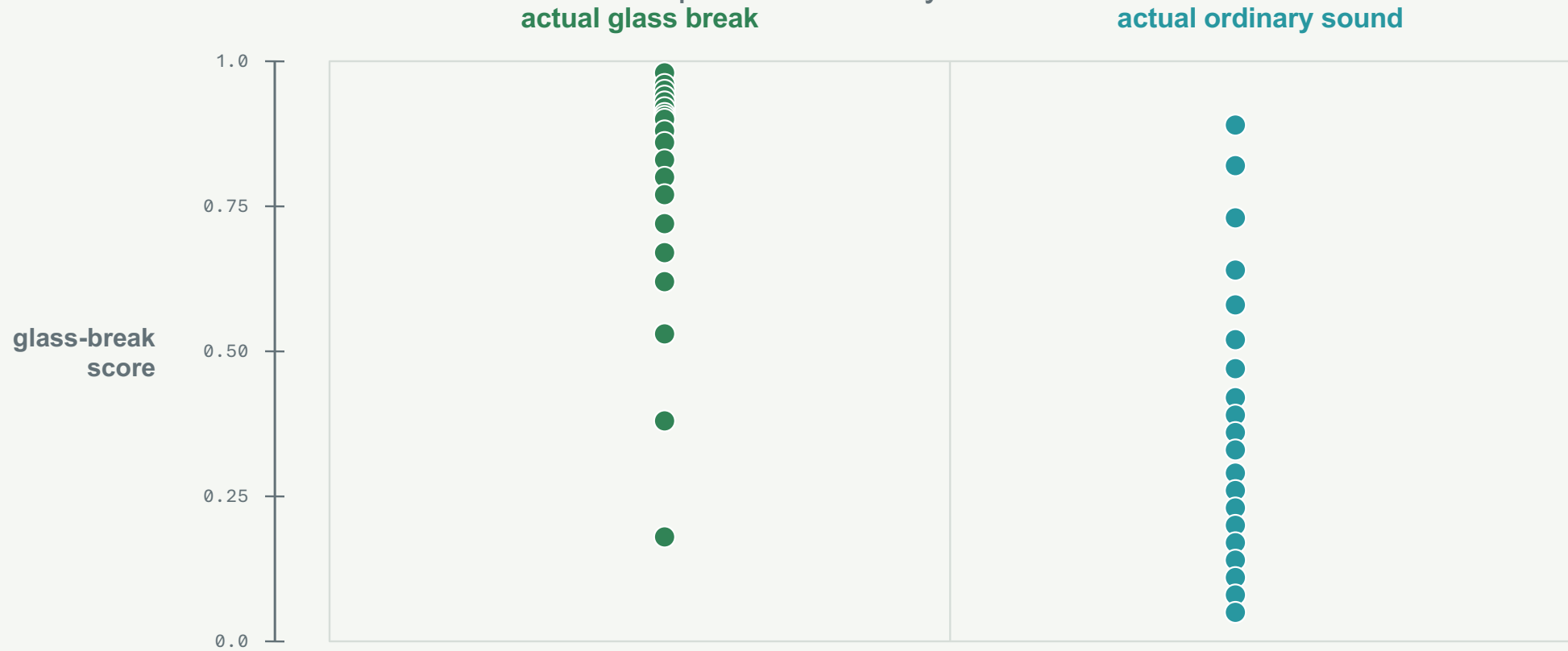


# Scores and Decisions

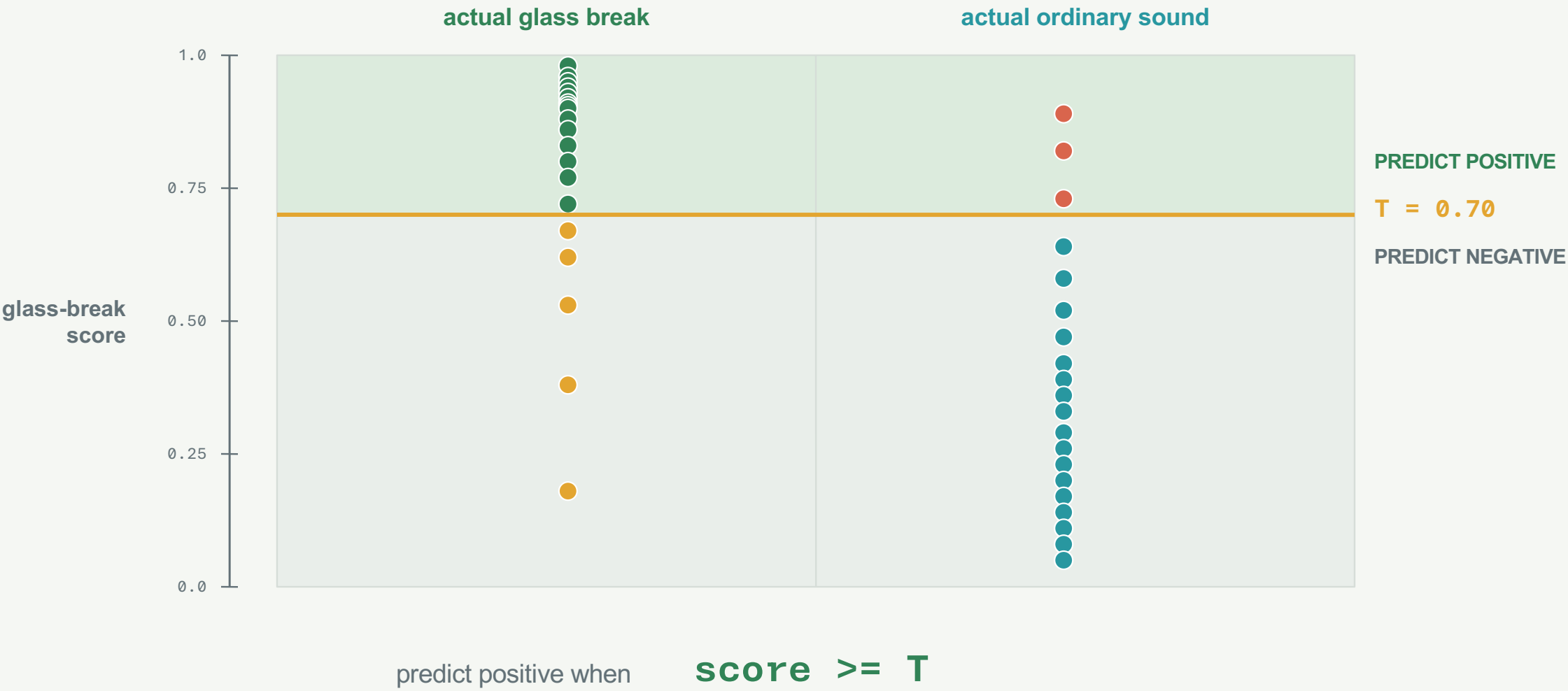


# Score Distributions

The classes overlap: some ordinary sounds score above some real breaks.



# Decision Threshold





# Application Consequences

## PHONE NOTIFICATION

A person reviews the alert  
before acting.

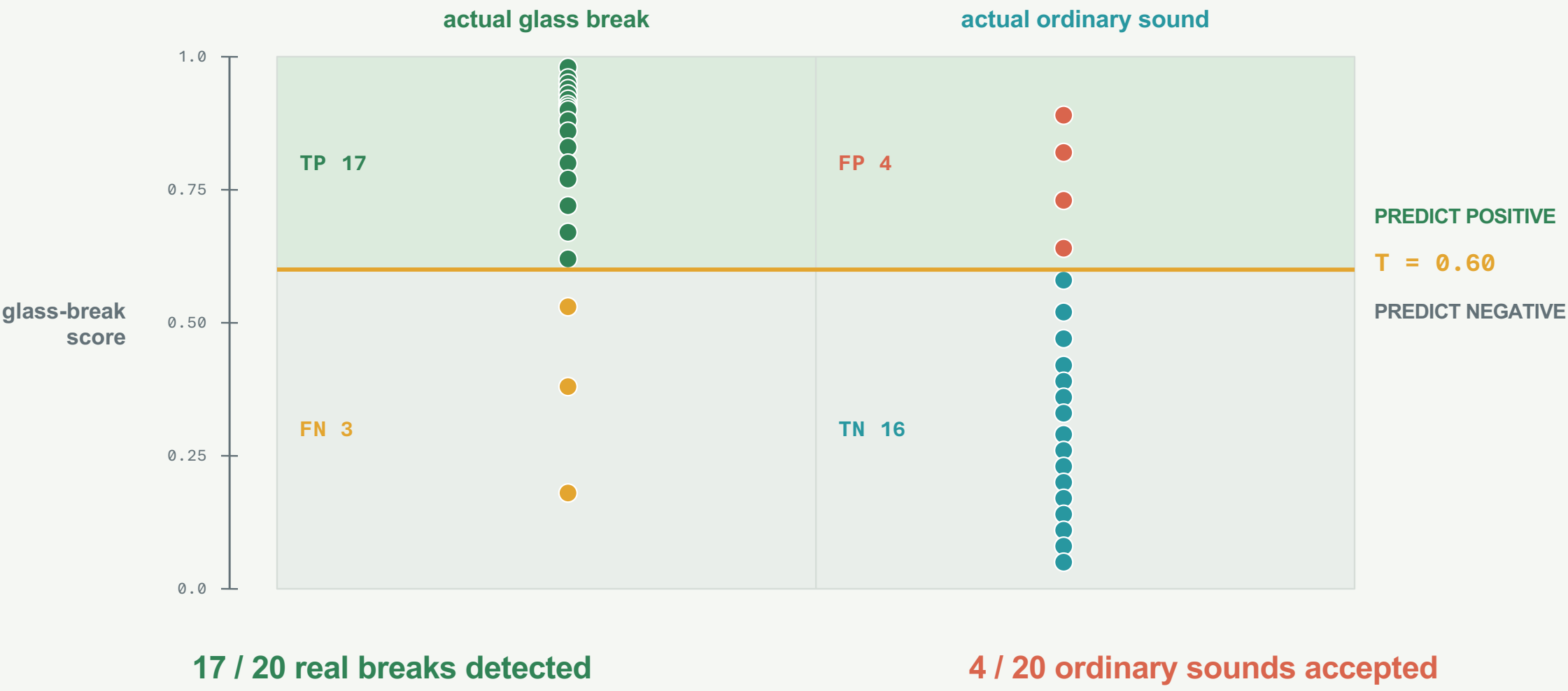
## AUTOMATIC RESPONSE

The system acts immediately  
without human review.

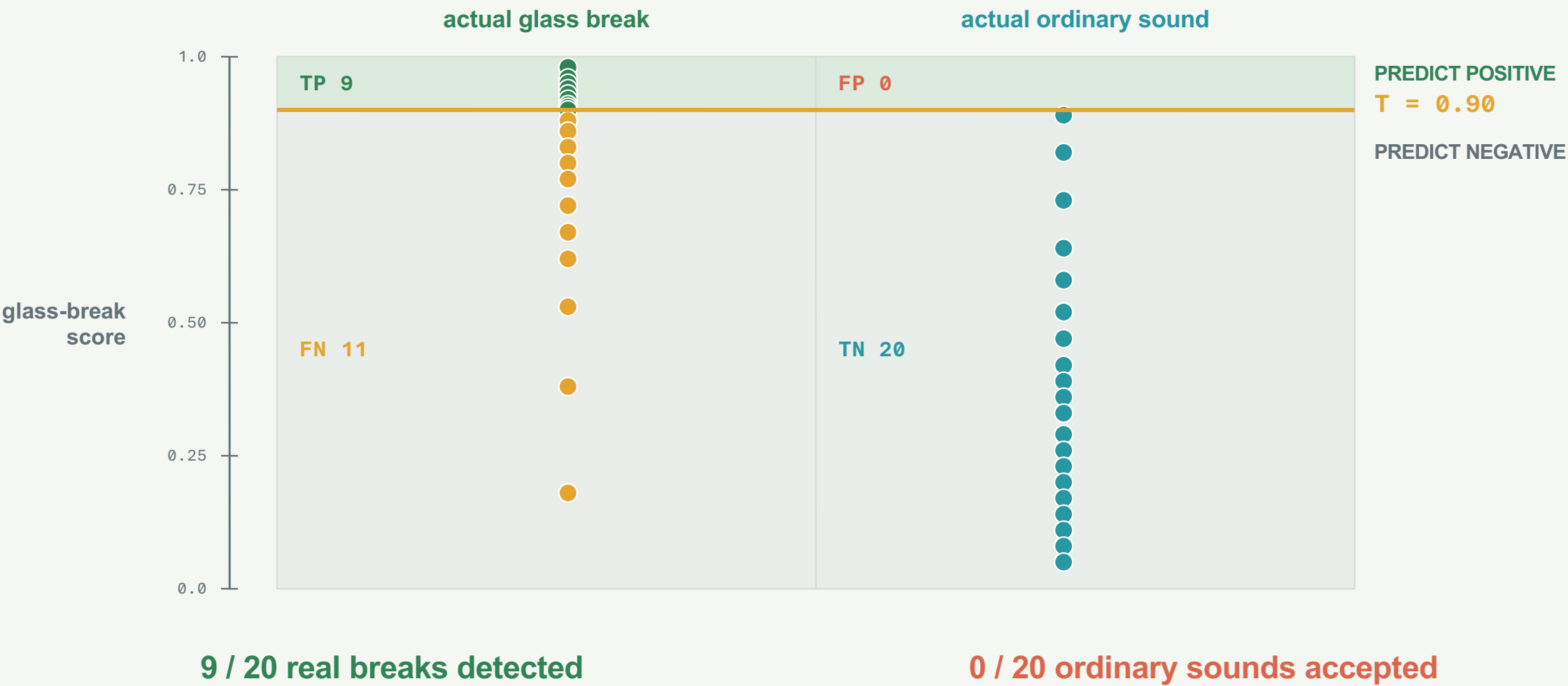
**Which error is more costly for each application?**

Talk with your neighbor before choosing a threshold.

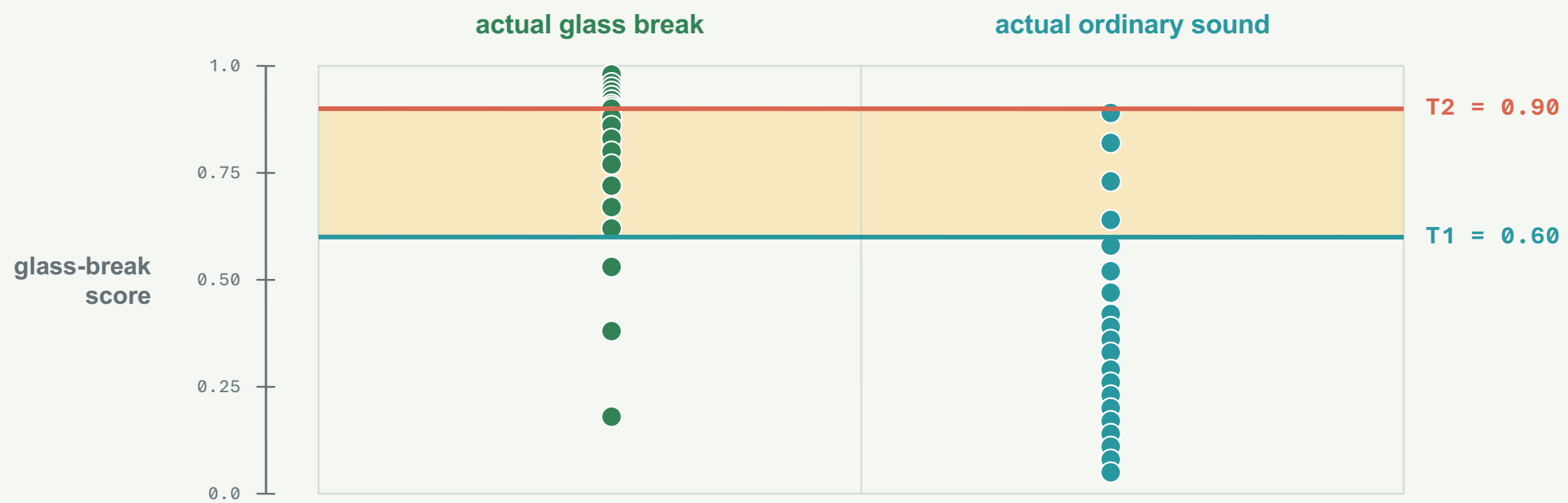
# Lower Decision Threshold



# Higher Decision Threshold



# Threshold Monotonicity

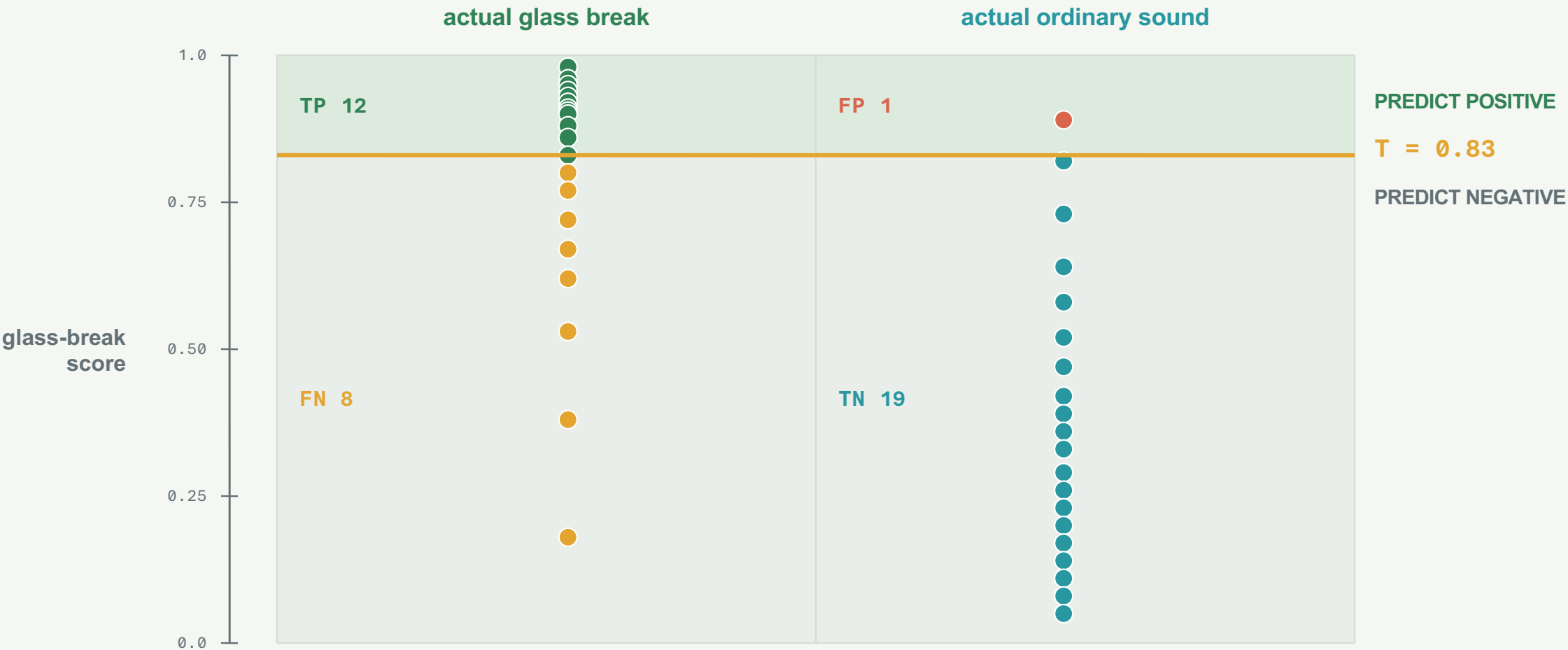


Raising the threshold moves the boundary upward; the accepted set can only shrink.

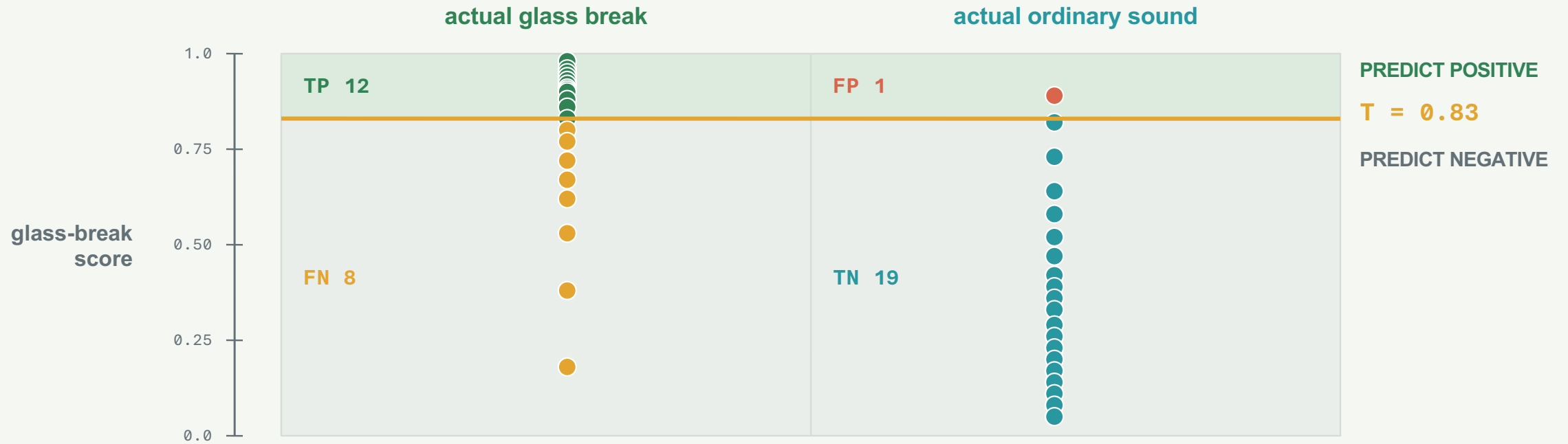
|                |                     |               |            |                                      |
|----------------|---------------------|---------------|------------|--------------------------------------|
| as T increases | predicted positives | TP and recall | FP and FPR | precision has no monotonic guarantee |
|                | 21 -> 9             | 17 -> 9       | 4 -> 0     |                                      |



# Binary Decision Outcomes



# Precision and Recall at One Threshold



among accepted predictions

$$\text{precision} = 12 / (12 + 1) = 0.92$$

How often is an accepted alert right?

among actual positives

$$\text{recall} = 12 / (12 + 8) = 0.60$$

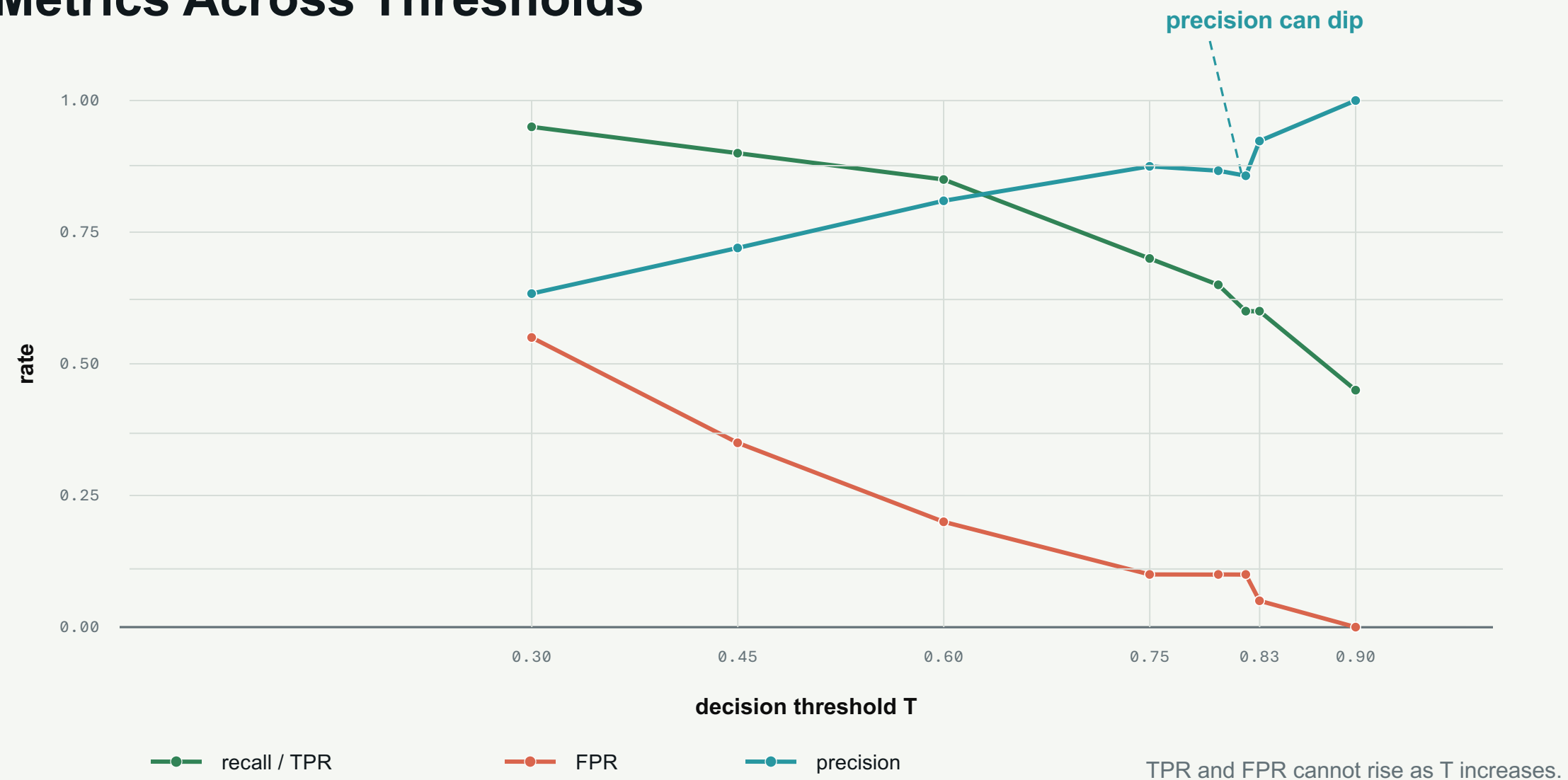
How often is a real break detected?

# Threshold Sweep

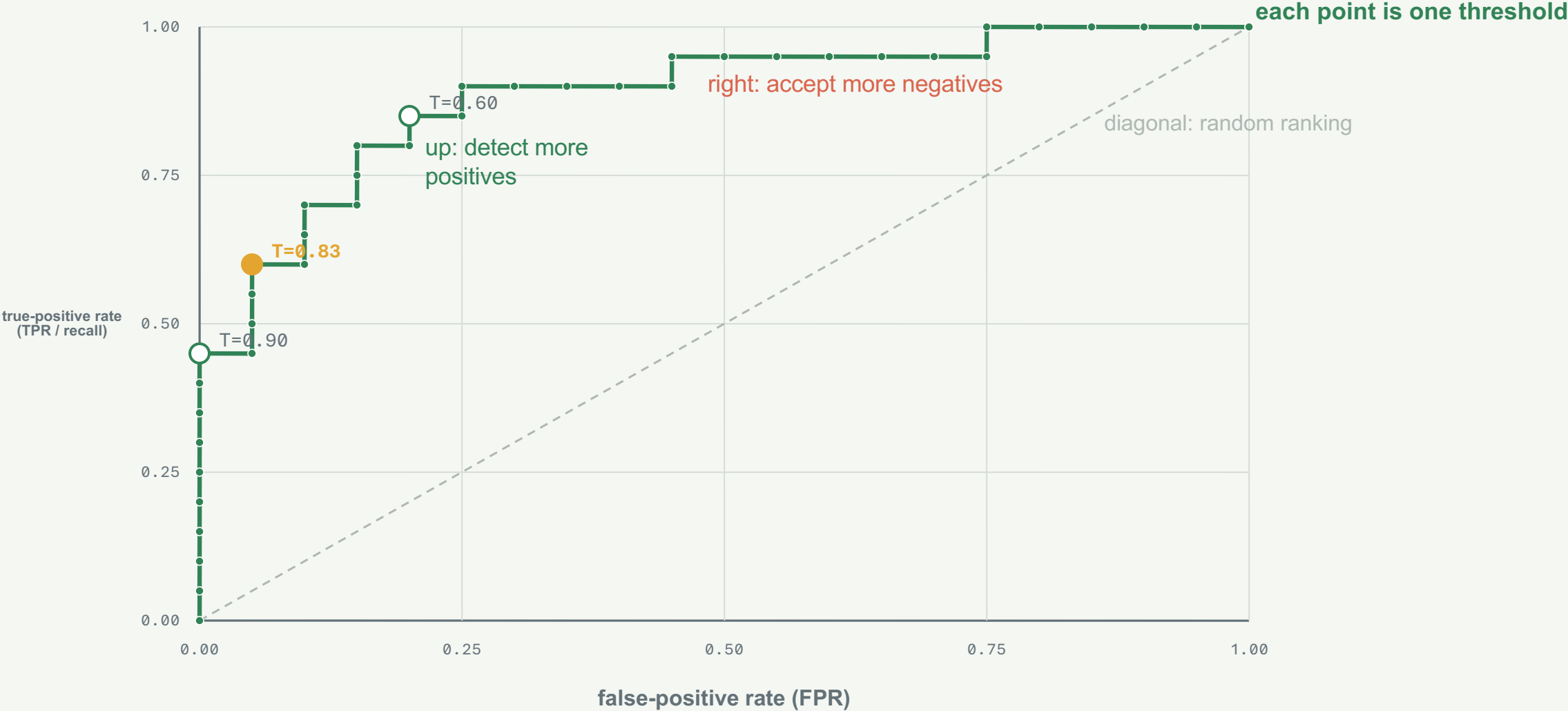
| threshold   | TP        | FP       | recall / TPR | FPR         | precision   |
|-------------|-----------|----------|--------------|-------------|-------------|
| 0.30        | 19        | 11       | 0.95         | 0.55        | 0.63        |
| 0.45        | 18        | 7        | 0.90         | 0.35        | 0.72        |
| 0.60        | 17        | 4        | 0.85         | 0.20        | 0.81        |
| 0.75        | 14        | 2        | 0.70         | 0.10        | 0.88        |
| 0.80        | 13        | 2        | 0.65         | 0.10        | 0.87        |
| 0.82        | 12        | 2        | 0.60         | 0.10        | 0.86        |
| <b>0.83</b> | <b>12</b> | <b>1</b> | <b>0.60</b>  | <b>0.05</b> | <b>0.92</b> |
| 0.90        | 9         | 0        | 0.45         | 0.00        | 1.00        |

same fixed model + same 40 labeled clips + different threshold

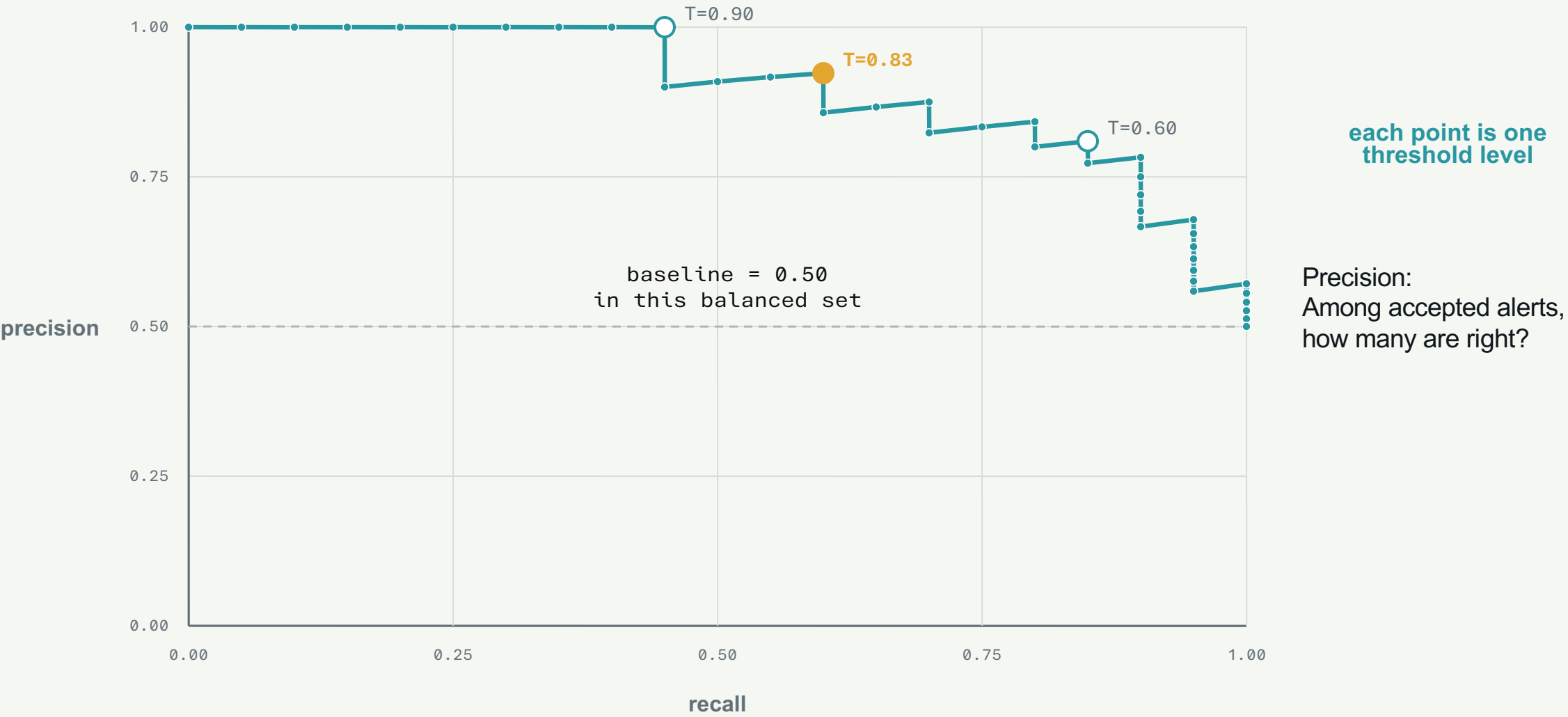
# Metrics Across Thresholds



# Receiver Operating Characteristic

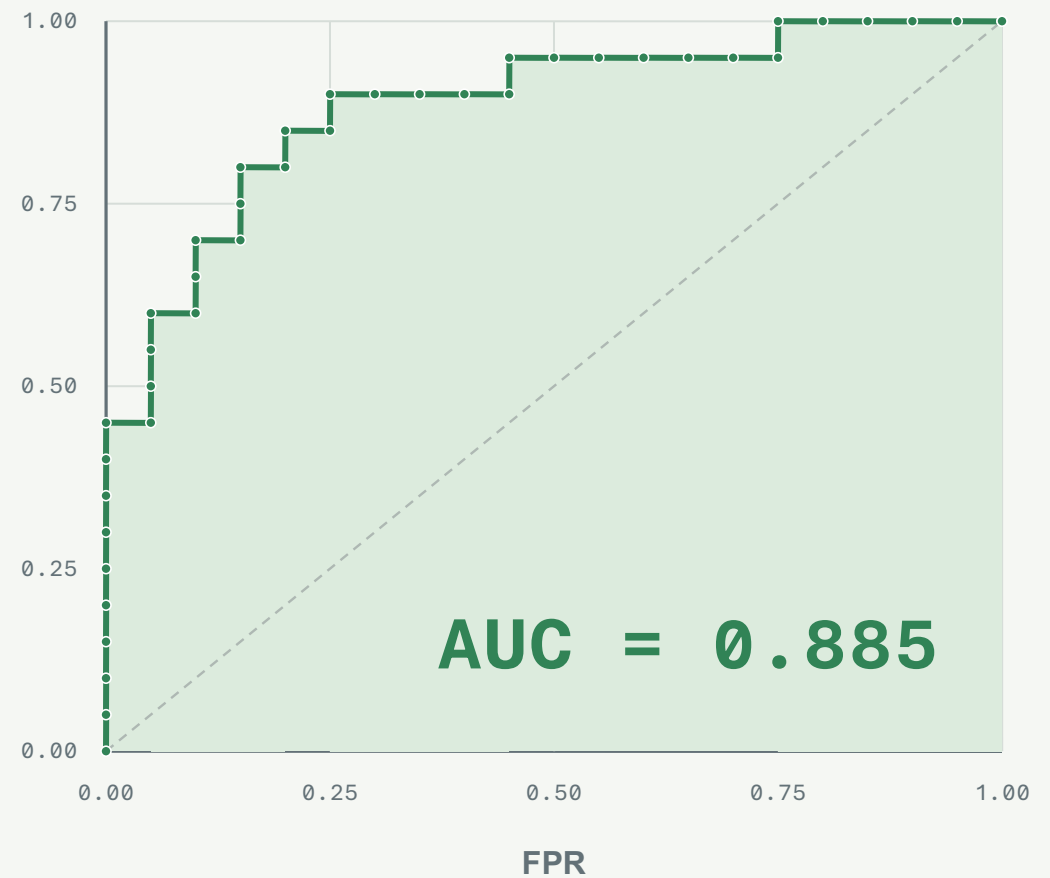
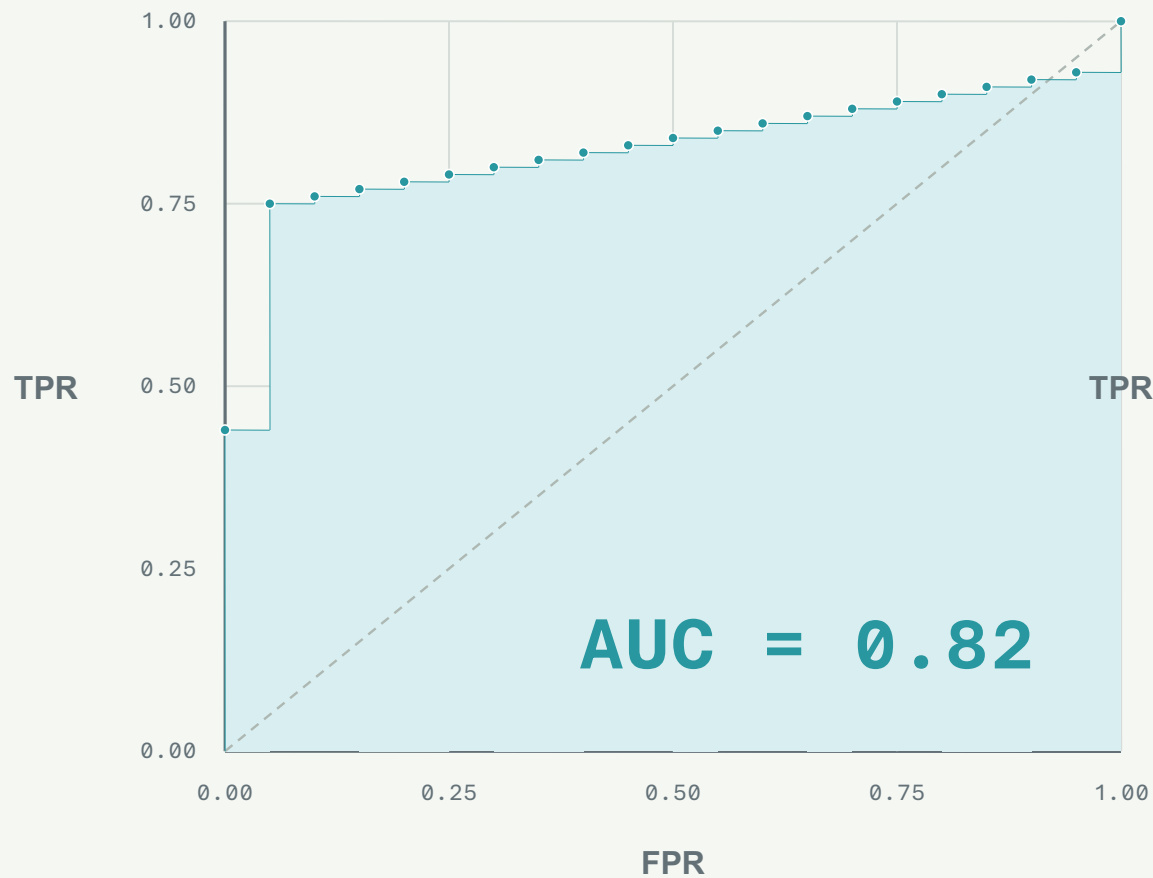


# Precision-Recall Curve



# Area Under the ROC Curve

$$AUC = \frac{354 \text{ correctly ordered positive-negative pairs}}{400 \text{ total pairs}} = 0.885$$





# Operating Requirements

Application goal: automatic response with limited false activations

example limit

**No more than 1 false  
activation**

in 20 ordinary validation clips



measurable constraint

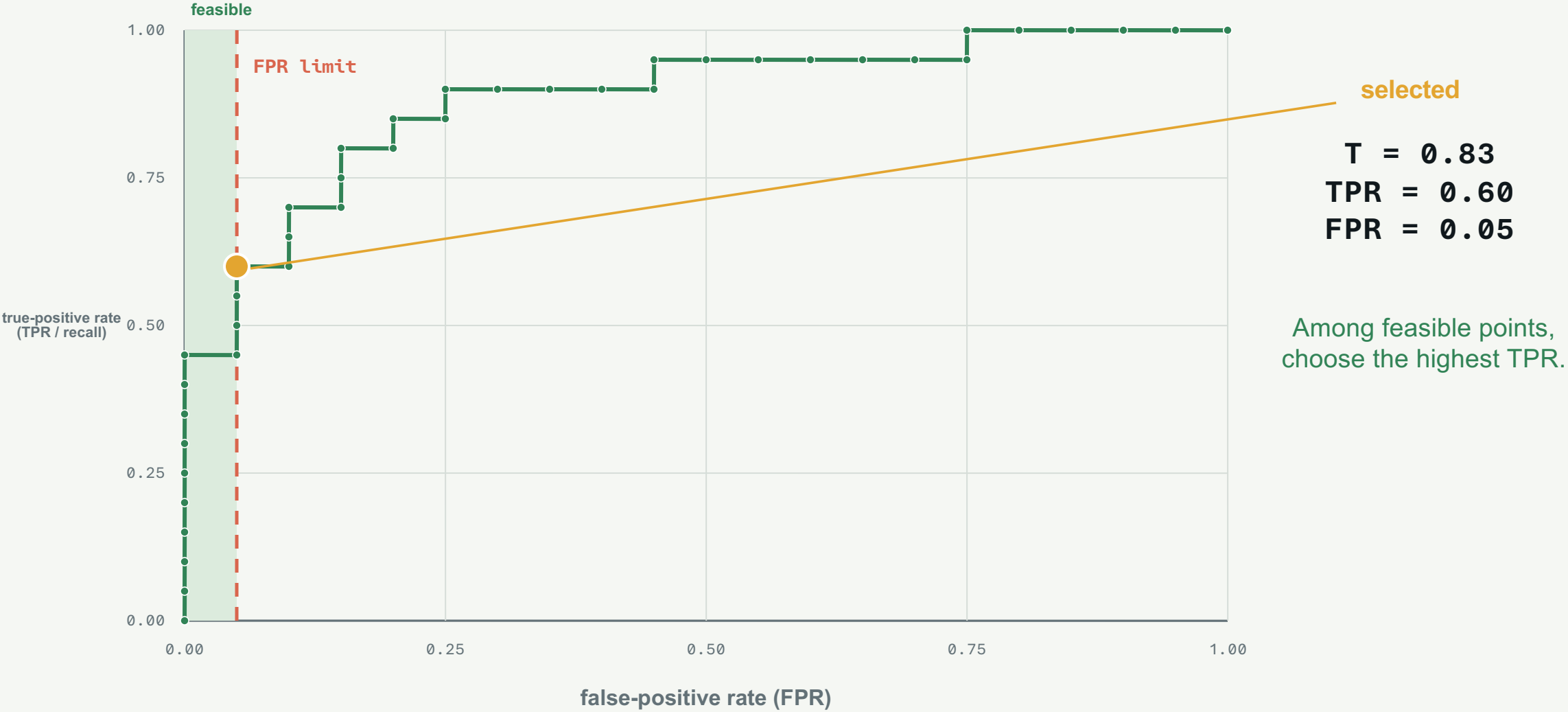
**$FPR(T) \leq 0.05$**

candidate thresholds must satisfy  
this limit

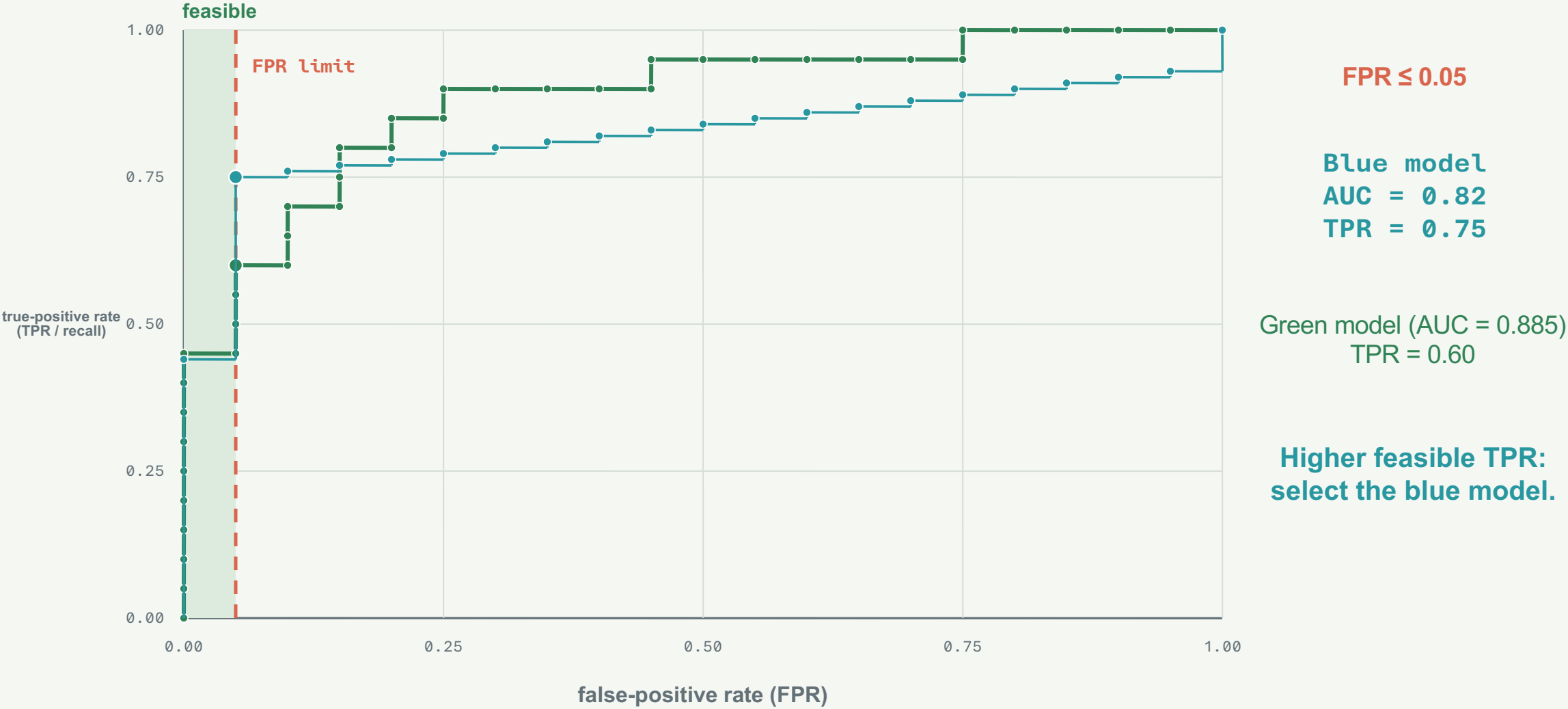
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**Choose  $T^*$  to maximize  $TPR(T)$  subject to  $FPR(T) \leq 0.05$**

# Constrained Operating Point Selection

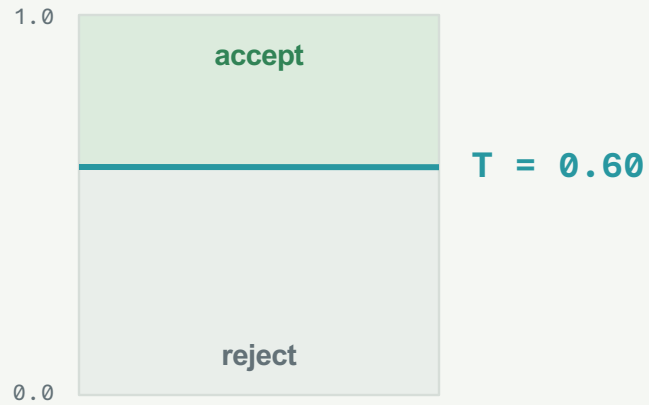


# Selecting Between Models at One Requirement



# Application-Specific Operating Points

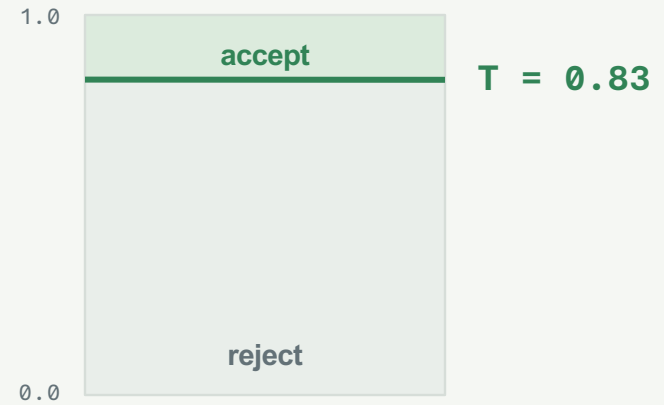
## PHONE NOTIFICATION



**TPR 0.85 | FPR 0.20**

More alerts; a person reviews them.

## AUTOMATIC RESPONSE



**TPR 0.60 | FPR 0.05**

Fewer false alarms; more real events missed.

**The model is identical. The operating policy is different.**

# Prevalence and Precision

same TPR = 0.60 and FPR = 0.05

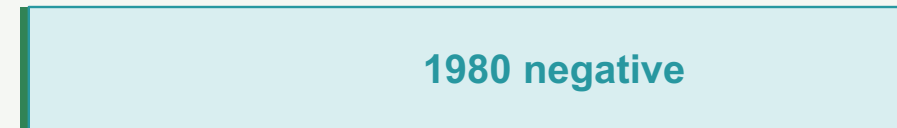
balanced evaluation set



12 TP + 1 FP

precision = 12 / 13 = 0.92

thought experiment: 1% positive windows



20 positive

12 TP + 99 FP

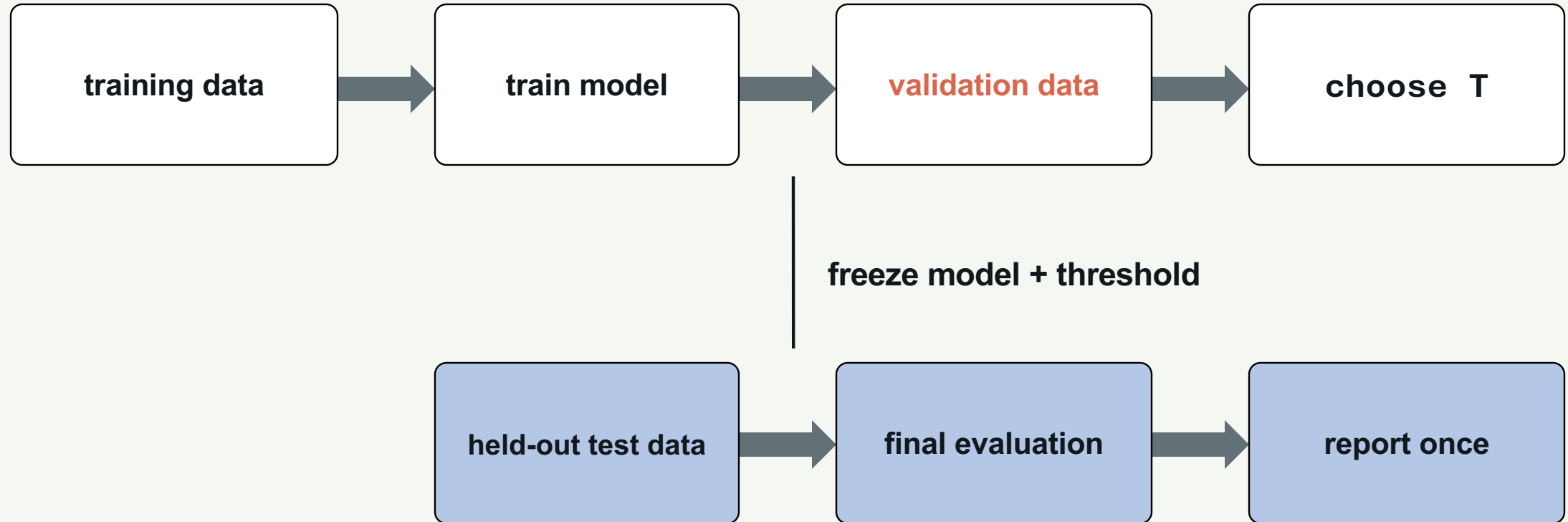
precision = 12 / 111 = 0.11

Window prevalence changed. The class-conditional rates did not.

Long background recordings are needed to estimate false alarms per hour.

# Threshold Selection Uses Validation Data

Note: Choose  $T$  on the training. Do not choose  $T$  on the final test set.



# Deployed Decision Policy

Changing any row creates a different evaluated policy.

|                     |                     |
|---------------------|---------------------|
| model version       | glass-audio-v3      |
| positive class      | glass break         |
| score               | glass-break output  |
| accept rule         | score $\geq$ 0.83   |
| validation evidence | TPR 0.60   FPR 0.05 |
| fallback            | no automatic action |